BT5240 - Computational Systems Biology
Assignment 2

Due date: March 11, 2026, 17:00 hrs

Instructions

If you need any assistance with computing, feel free to approach the TAs. Evaluation will be based on the code(s), the answers, and the methodology.

Academic Integrity: Discussing problems verbally with friends is allowed, but copying or looking at codes is not permitted (use of ChatGPT/LLMs is also NOT permitted). Transgressions are easy to find and will be reported and dealt with strictly. Mention any collaboration (discussions only!) in your solutions. Late submission penalties: 1 second to 24 h: 20%; 24 to 48 h: 40%; >48h: 60% Early submission bonuses: >24h: 10%, >48h: 20%

Submission

Join the class on Turnitin using class code **52127280** and enrollment key *bt5240_2026*. You need to make two submissions: a Python file and a written report.

1. Code guidelines:
    a. Submit a single *.py* file containing the code for all the questions.
    b. Ensure the code is clean and properly formatted.
    c. Use clear comments to explain logic when necessary.

2. Report guidelines:
    a. Prepare a report that includes answers to all the questions, all required plots/tables properly labeled, clear explanation of the results. For question 3, include step-by-step procedure followed in Cytoscape. Submit a single PDF file.
    b. Use Times New Roman, font size 12, 1.5 line spacing, with clean and consistent formatting.

Question 1:

Download the [yeast protein–protein interaction] network and load it into NetworkX.

a) Report basic network statistics, including the number of nodes and edges, density, and diameter.
b) Compute Degree, Betweenness, and Closeness centrality and briefly explain what each measure represents in the context of a protein–protein interaction network.
c) Identify the top 5% of nodes for each centrality measure and treat them as putative essential proteins.
d) Independently remove each of these node sets from the network and measure the size of the largest connected component.

e) To assess statistical significance, simulate 100 iterations of random removal of 5% of nodes and compare the results with the centrality-based removals by computing Z-scores. Comment on the suitability of using Z-scores based on the distribution obtained from random removals, and conclude which centrality measure best supports the centrality–lethality hypothesis for this network.

f) Find at least three nodes with high Betweenness Centrality but relatively low Degree. Based on PPI network topology, explain why these specific "bottleneck" proteins might be biologically essential even if they don't have many neighbours.

g) Using a database like SGD (Saccharomyces Genome Database), look up the top 2 nodes from your Closeness Centrality list. Do their known biological functions justify their **central** position, or could their high centrality be an artefact of study bias?

Question 2:

Using the Zachary Karate Club Network, which represents friendships among members of a karate club that eventually split into factions, compare the Girvan–Newman algorithm and the greedy modularity maximization algorithm.

1. Use NetworkX to apply greedy modularity maximization while restricting the maximum number of communities to 2, 3, and 4. For each case, report the modularity value and community sizes. Examine how modularity changes with increasing number of communities, identify the partition with maximum modularity, and determine whether it recovers the known real-life split.

2. Use NetworkX to apply the Girvan–Newman algorithm and analyze the first five successive partitions produced by the Girvan–Newman algorithm. For each partition, record the number of communities, community sizes, and modularity. Identify the partition with maximum modularity, check whether it aligns with the real-life split, and describe how modularity changes as edges are progressively removed.

3. Compare the two algorithms in terms of maximum modularity achieved, computational complexity, and their ability to detect the real-life community structure.

4. In the Girvan-Newman algorithm, multiple edges often have the exact same maximum edge-betweenness. Explain how the NetworkX function handles these ties.

5. The greedy modularity maximization algorithm is known to suffer from a "resolution limit." Did you observe any instances where the algorithm merged small, distinct functional groups into a single community to artificially boost the modularity score?

Note: Load the network using NetworkX. Real split can be obtained from node attributes.

Question 3:

Microbial association networks from two environments (i) Hospital and (ii) Metro stations are provided as edge lists with normalized correlation values as edge weights. Using Cytoscape, analyze each network individually and compare them.

1. Plot the degree distribution.
2. Plot the average clustering coefficient as a function of node degree.
3. Compute the average node degree, average clustering coefficient, and characteristic path length for each network and comment on the observed differences.
4. Identify associations unique to each environment as well as those common to both networks using Cytoscape. For the common interactions, examine how the normalized correlation values differ between environments and visualize these changes using an appropriate plot. Finally, identify the top five most interacting microbes in each environment from both the original networks and the unique interaction networks, and comment on your observations.